Joshua Prettyman, Ph.D.

Data Scientist, Mathematics Ph.D., Python Developer

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Personal Statement

I built an ML-powered SaaS platform that increased SEO productivity by 20×, taking Blink SEO's internal tooling from spreadsheets to a marketable product now used by multiple agencies: *Macaroni Software*. I created the role after being hired as a one-man data/dev team with the brief: 'use data to improve processes'.
My research at the National Physical Laboratory produced a novel scaling indicator for tipping point detection in multidimensional time series, published in Environmental Research Letters, 2022.
I am seeking a remote data science role where I can build production systems and continue solving complex technical problems.

Technical Skills

- Regular use of Python libraries: ScikitLearn, Numpy, Pandas, NLTK, Matplotlib, Plotly, PyTorch.
- Python development: Data Science/ML, async, package management, REST, GraphQL.
- Full stack development with Bash, Python, Google Cloud Platform, SQL and JavaScript.
- Huggingface transformers and LLMs, ollama LLM, OpenAI Assistant API.
- Experience with C++, MATLAB, Retool, data engineering, dashboards, scientific computing.

Professional Experience

Data Scientist — Blink SEO / Macaroni Software *Nov. 2021 to Dec. 2024*

Built a full-stack ML platform from scratch, increasing team productivity by 20× and enabling year-long SEO campaigns to be delivered in weeks.

- Worked with the team to replicate and then improve their (spreadsheet-based) processes with continual feedback and deployment.
- Macaroni does the (typically manual) data-engineering for SEO by aggregating data from various sources, streamed daily (or on request) to a *BigQuery* database via the python backend (*GCP Compute Engine*: Linux/Debian) which utilises various REST APIs and preprocessing steps.
- Client onboarding, data imports, data analysis tasks and ML-assisted recommendations can be triggered in the app and run asynchronously in the backend using a (*PostgreSQL*-based) job-queue system which I wrote for this purpose.
- Quantitative data are used for various visualisations in the app, powered by *Plotly*.
- Text data (on-page content and search terms) are processed and combined with quantitative data (clicks, impressions, etc.) to surface actionable recommendations for new content and site taxonomy. This uses NLP and Clustering algorithms (*NLTK*, *huggingface*, *ScikitLearn*) but also a deep understanding of SEO, necessitating regular exchanges with the SEO delivery team and frequent iterations.
- I also integrated an LLM (*huggingface*) to generate ready-to-go suggestions for content gaps, allowing customers to see immediate impact.
- Frequent, one-off data tasks for the delivery and management teams including data engineering, content generation, and providing analyses and visualisations for clients and investors.

Data Science Researcher — National Physical Laboratory (NPL) *Sep. 2015 to Feb. 2021*

Produced a novel scaling indicator for tipping point detection in multidimensional time series.

- Presented results for stakeholder meetings at NLP, in published articles, and at international conferences.
- Developed methods to detect tipping points in dynamical systems, working with large time series datasets and building stochastic models to test hypotheses.
- Code written in *MATLAB* and *Python*. Visualisations produced using *Matplotlib* and *XMGrace*.
- Accessing, cleaning, interpolating and correctly interpreting large, raw meteorological datasets.

Associate Lecturer — Sheffield Hallam University *Sep. 2017 to Jul. 2019*

Taught mathematics, statistics and computing courses from Foundation Degree to Masters.

- Taught a course on *Microsoft Excel for Business.*
- Responsibilities included lesson preparation, course planning, marking, leading tutorials.
- Manned the "Maths Help" desk three times a week, providing support to students of all departments.

Informatics developer (internship) — UK Met Office Informatics Lab *Jun. to Aug. 2017*

A three-month project assessing the viability of using live data from amateur meteorologists to augment official observation data.

- Accessed internal and external data via APIs using Java.
- Data cleaning and analyses in Python.
- Worked as part of a team and participated in daily stand-ups.

Academic background

Ph.D. Mathematics — University of Reading
Thesis: *Tipping Points and Early Warning Signals with Applications to Geophysical Data.*

- Thesis develops spectral analysis methods to detect correlations in multidimensional time series, which are used as Early Warning Signals for tipping events in stochastic dynamical systems.
- Publication of three papers in respected journals:
  - *A novel scaling indicator of early warning signals* (EPL, 2018)
  - *Generalized early warning signals in multivariate and gridded data* (Chaos, 2019)
  - *Power spectrum scaling as a measure of critical slowing down* (ERL, 2022)
- Implemented mathematically derived methods numerically by discretising and linearising.
- Participation in several international conferences and workshops.
- Three-month data-focussed internship at the UK Met Office (see *Professional Exp.* section).

M.Res. Mathematics — Imperial College London (*Distinction*)
Dissertation: *Mesh Generation Using Solutions of an Optimal Transport Problem.*

- Dissertation uses solutions to a linearisation of the Monge-Ampère equation to generate an adaptive, optimally distributed mesh for numerical models.
- Working in C++; monitoring CPU usage and convergence time over various scenarios.
- Participation in weekly soft skills workshops and monthly internal showcases.
- Attendance at weekly Mathematics of Planet Earth seminars hosted by the Centre for Doctoral Training.
- Completed taught courses in Python, R, Dynamical Systems, Stochastic differential equations, Bayesian Analysis and Probability.

MA Mathematics — University of Edinburgh (*First Class Hons.*)
Dissertation: *Growth and its Applications in Graph Theory.*

- Dissertation explores patterns in paths through connected digraphs.
- For the dissertation project I wrote a program with a simple UI to investigate the properties of these digraphs: this then became a teaching aid, earning a letter of thanks from the principal.
- Requisite taught courses for the Pure Mathematics degree besides optional modules in computing and mathematics education.